

LLD Practice Platform

Engineering assignment submission

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This pack contains the research note, the design note, the README and the AI usage note. The working prototype and its tests are in the repository.

Research note

What the learner problem actually is, what already exists, where the gaps are, and the product direction that follows.

1. The learner problem

LLD practice has an unusual shape: it is **easy to start and almost impossible to self-assess**.

A learner can read "design a parking lot", spend forty minutes producing a class diagram, and come away with no idea whether it was any good. Three properties of the domain cause this:

There is no single correct answer. Unlike a DSA problem, where a test suite settles it, a parking lot can put pricing in a `PricingStrategy`, a `RateCard`, or on `Ticket` itself and be defensible in all three. So the obvious feedback mechanism, compare against a reference solution, is actively wrong. It teaches conformity to one author's taste and punishes good designs that differ.

The mistakes are invisible from the inside. The errors that matter are structural: a class doing two jobs, cardinality that cannot express a real case, a switch statement that will be reopened for every new requirement. A learner who makes one of these does not experience an error. Their design looks finished to them. This is precisely the gap a reviewer closes and self-study cannot.

The graded artefact is partly an argument, not an artefact. In a real interview, roughly half the signal is the reasoning: *why this shape and not the other one, what does your choice make harder*. A learner practising alone produces the diagram and skips the argument entirely, because nothing forces it out of them.

The consequence is a plateau. Learners solve Parking Lot, Elevator and Vending Machine, feel no more confident than before, and conclude LLD is a thing you either have a knack for or do not.

2. What already exists

Approach	What it does well	Where it leaves the learner
Blog posts & GitHub repos (low-level-design-primer , awesome-lld lists)	Free, plentiful, often genuinely good reference solutions.	Purely read-only. You compare your answer to theirs and see a difference, with no way to know whether the difference matters. No practice loop at all.
Books & courses (Head First Design Patterns, Grokking the OOD Interview)	Excellent at teaching the vocabulary: patterns, SOLID, when to use which.	Worked examples, not exercises. The reader is never asked to produce a design that is then examined. Knowledge without a feedback signal.
DSA platforms (LeetCode, HackerRank)	The loop is perfect: attempt → automatic verdict → retry, with history.	The verdict comes from test execution, which LLD has no equivalent of. Their LLD-adjacent content is multiple-choice or "here's the answer, compare yourself".
Mock interview marketplaces (Pramp, interviewing.io)	The highest-quality feedback available: a human who probes the reasoning.	Expensive, scheduled, and roughly one problem per session. Practising a design five times to see it improve is not economically possible.
General-purpose LLMs (paste your design into ChatGPT)	Free, instant, surprisingly perceptive on structure.	Three failure modes seen consistently in testing: it is agreeable, praising weak designs; it drifts, giving different verdicts for the same input; and it grades against whatever reference solution it happens to recall, marking down unconventional-but-sound answers. It also never says "you did not mention pricing at all" reliably, because it has no fixed checklist.

3. The gaps worth building into

Gap 1: Nothing closes the loop. Reading material is read-only; mock interviews do not repeat cheaply. Nobody offers *attempt → feedback → attempt again → see the difference* for LLD, which is exactly the loop that made DSA practice work.

Gap 2: Feedback is either rigorous or scalable, never both. A human reviewer is rigorous and does not scale. A raw LLM scales and is not rigorous; it will not reliably notice a missing requirement, and it varies run to run.

Gap 3: Learners are not made to state the parts that get graded. A blank text box lets someone write three fluent paragraphs while never naming a class's responsibility or a relationship's cardinality. The submission format itself is a teaching tool that everyone leaves on the table.

Gap 4: Nothing distinguishes a fact from an opinion. When an LLM says "your Vehicle class violates SRP", the learner cannot tell whether that is checkable or a guess, so they either accept everything or trust nothing.

4. Product direction

DesignDojo: a repeatable practice loop for LLD, with feedback that shows its work.

Four positions follow directly from the gaps above.

1. The submission is structured, not prose. The learner names their classes, writes one line of responsibility for each, and states relationships *with cardinality*. This is the product's main teaching device: it makes the skipped decisions unskippable, and it happens to give the evaluator something precise to read. Code and Mermaid class diagrams are accepted too, for learners who think in those, all three normalise to the same internal model. (*Gap 3.*)

2. Evaluation is deliberately split in two. Checkable things (is each stated requirement addressed, is any class carrying twelve members, is cardinality stated, does the design mention what happens when the lot is full) are decided by reproducible rules. Judgement calls (is this abstraction well chosen, does the reasoning hold) go to a language model, which is given the rule findings and told not to duplicate them. The rubric decides per dimension how much each source is trusted. (*Gap 2.*)

3. Every finding is labelled with its source, on screen. A learner can see that "nothing owns pricing" came from a rule that will show its matched evidence, while "Vehicle is doing two jobs" is a reviewer's opinion they are free to argue with. Requirement coverage displays the learner's own words that matched, so a wrong verdict is arguable rather than authoritative. (*Gap 4.*)

4. Attempts are never overwritten. Every attempt is kept, numbered, and charted. The product's core screen is not the score; it is the second attempt scoring higher than the first on the dimension the learner was told to fix. (*Gap 1.*)

What is deliberately not in the MVP

Accounts and auth (a per-browser learner id is enough to make history real), a problem-authoring UI (problems are seeded data: a mentor could add one without an engineer, which is the point), social or leaderboard features (they reward score-chasing over the retry loop), and rendered diagram editing (Mermaid text diffs between attempts; a canvas does not).

How this would be validated

The metric that matters is **score improvement between attempt 1 and attempt 2 on the same problem**: the loop either works or it does not. Secondary: what share of learners submit a second attempt at all, and whether the dimension a learner was told to fix is the one that moves.

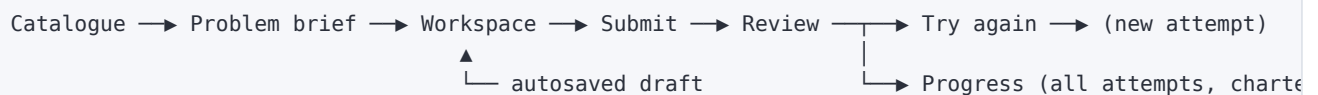
Design note

The MVP, the user flow, the classes that carry the domain, how evaluation works, and what was traded away.

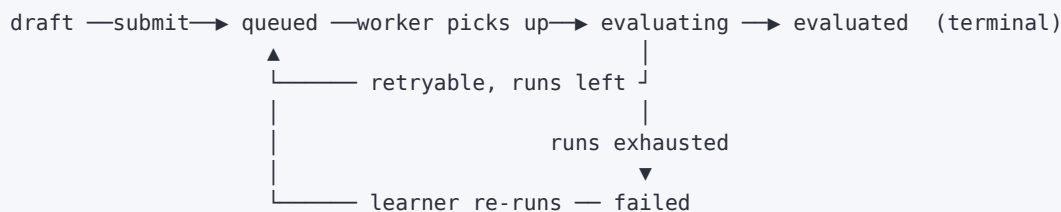
1. The MVP in one paragraph

A learner picks a problem from a seeded catalogue, works on a design in one of three formats, and submits it. Submission returns immediately with the attempt `queued`; a background worker normalises the submission into a common design model, runs it through an evaluation pipeline (deterministic rules always, a language model when available), and stores a scored review. The client polls until the attempt leaves a pending state, then shows the score per rubric dimension, the feedback with each item labelled by its source, and requirement-by-requirement coverage with the matched evidence. Every attempt is kept, so the history view charts score across attempts per problem.

2. User flow



The states behind it are the attempt's own, not a spinner:



3. The domain

Three aggregates, one of them doing most of the work.

Problem : authored content, immutable

Carries the learner-facing brief *and* the grading key: `requirements` (each with `signals`, the synonym set that counts as addressing it), `expectedConcepts` (with aliases and a *why*), `pitfalls` (problem-specific traps with trigger terms), and a `Rubric`. The DTO layer strips the key before it reaches the client.

The design decision here is that a problem is data, not code. Adding "Design a Chess Game" is a new entry in `seedProblems()`: the deterministic checker picks up its signals, concepts and pitfalls with no new classes. Problem authoring is a content task.

Rubric : six fixed dimensions, per-problem weights

`requirement_coverage`, `abstraction_quality`, `relationships`, `extensibility`, `edge_cases`, `tradeoff_reasoning`.

The axes are fixed deliberately: a learner comparing attempt #1 to attempt #4 needs them to mean the same thing every time. Problems tune the weights (Elevator leans on relationships and trade-offs; Vending Machine on edge cases).

Each dimension also carries a `deterministicShare` $\in [0,1]$: how much of that dimension's score comes from rules versus the model. This one number is where the platform's whole evaluation philosophy lives, expressed as data rather than as branching inside the evaluator:

Dimension	Deterministic share	Why
Requirement coverage	0.7	A finite authored checklist. Mostly a fact.
Relationships	0.5	Cardinality present/absent is a fact; whether it is <i>right</i> is judgement.
Edge cases	0.4	Vocabulary is detectable; sufficiency is not.
Abstraction quality	0.3	Crowded classes are detectable; good taste is not.
Extensibility	0.3	Seams are detectable; whether they are the <i>right</i> seams is not.
Trade-off reasoning	0.0	Only a reader can judge an argument.

Attempt : the aggregate root of the practice loop

Owns its status, its submission, its evaluation, its event timeline and its run count. **Every state change goes through a method on this class, and every method consults a single transition table** (`AttemptStatus.ts`). Nothing outside the class assigns to `status`, so "can an evaluated attempt be resubmitted?" has exactly one answer in exactly one file.

Two behaviours worth calling out:

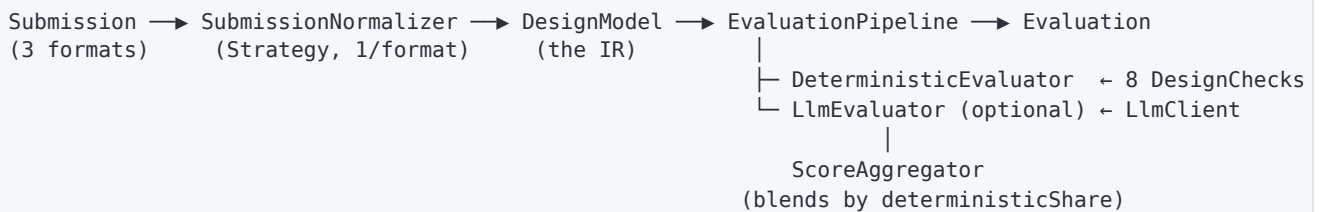
- `submit()` **validates before it transitions**. A submission too thin to be worth evaluating leaves the attempt untouched and still editable: nothing the learner typed is lost. (*This ordering was a real bug, caught by a test; see `AI_USAGE.md`.*)
- `failEvaluation(reason, retryable)` decides between re-queueing and giving up, based on the retryability of the error and the runs remaining. The attempt, not the worker, owns its own retry budget.

Submission : a discriminated union, not a hierarchy

`StructuredSubmission` | `CodeSubmission` | `DiagramSubmission`. Inert data that crosses the wire and the database; behaviour lives in the normalisers that read it.

The product opinion in the MVP is that the structured format is primary. Free prose is pleasant to write, impossible to give precise feedback on, and lets a learner skip naming responsibilities and cardinality: the very things being taught. Code and diagram exist for learners who think in them.

4. The evaluation architecture



DesignModel : the intermediate representation

Entities, relationships, operations, narrative, plus a flattened lower-case `searchText` for signal matching. **Evaluators depend on this and never on the raw submission.** That is what makes the format axis and the evaluator axis independent: a new format is one normaliser and zero evaluator changes; a new evaluator works across all three formats for free.

DesignCheck : the deterministic half

Eight small, independent, synchronous rules: requirement coverage, concept coverage, responsibility distribution, relationship clarity, extensibility seams, edge-case awareness, trade-off articulation, known pitfalls. Each returns a score for one dimension, a rationale, and feedback items.

Handling "more than one valid answer" is a matter of wording and of floors. `ConceptCoverageCheck` never says a concept is missing, it says nothing was *detected* that owns it, and asks whether that was deliberate. It has a score floor, so disagreeing with the reference costs some marks and never all of them. Requirement coverage, where there genuinely is a right answer, has no such floor.

Three refinements came out of running hand-written designs through the platform and finding the verdicts unconvincing:

Concepts are scored three ways, not two. A concept can be *dedicated* (a type is named for it: full credit), *folded* (it appears only inside another type's members, e.g. `calculateFee()` on a god class, half credit and a question), or *absent* (no credit). Grading `folded` the same as `dedicated` let a design that merely name-drops score as well as one that models the concept; grading it the same as `absent` would assert that the reference decomposition is the only correct one. Half credit says what is actually true, the concept is present, but its ownership is unclear from the design alone.

Edge-case awareness is scored from the problem's own implied requirements, not a global word list. Every problem authors the cases the brief deliberately omits (a full lot, two cars racing for the last spot, an equal split of 100 across three people) each with tuned signals. Those are a far sharper instrument than generic failure vocabulary, and they reward noticing the case rather than using a particular word. The generic vocabulary stays as a smaller secondary signal, since a design can be thoughtful in ways the problem's author did not anticipate. Reading the implied requirements here as well as in `RequirementCoverageCheck` is deliberate rather than double counting: that check asks *did you cover the brief*, this one asks *did you think past it*.

A check that finds nothing abstains rather than scoring 100. `PitfallCheck` returns no score at all when a design walks into none of the known traps. Not falling into a handful of named pitfalls is the absence of evidence, not evidence of good abstraction, and scoring it as a perfect result quietly lifted every design's

abstraction average for doing nothing.

This half is the platform's **floor**: dependency-free, single-digit milliseconds, and reproducible. Whatever happens to the model, a learner who submits gets a real score with real evidence.

LlmEvaluator : the qualitative half

Judges only the five dimensions where taste is required, and is **given the deterministic findings in its prompt and told not to repeat them**, so the two halves compose into one review rather than two overlapping ones. Output is strict JSON validated by a Zod schema; the parser scans for the first brace-balanced object (ignoring braces inside strings) because models wrap JSON in prose. Temperature 0.1: the same design resubmitted should get the same review, or the score history means nothing.

Three prompt rules exist because of observed behaviour without them: without *"there is more than one correct design"* it graded by similarity to the reference; without *"quote something specific from their submission"* it produced fluent generic advice; without *"do not comment on requirement coverage"* it duplicated and sometimes contradicted the rules.

ScoreAggregator

Blends per dimension by `deterministicShare`; falls back to whichever source reported when one is missing; **renormalises the overall score over the dimensions actually assessed**, so a degraded run is not silently scored out of a different total.

5. What happens when evaluation is slow or fails

Three levels, and at no level is the learner's submission lost or the error swallowed:

1. **An optional evaluator fails** (the usual case: model rate-limited, timed out, or returned unparseable JSON). The pipeline completes with a deterministic-only review, flags `degraded: true` with the reason, and the UI shows a "Partial review" banner with a re-run button. The learner gets a genuinely useful review, clearly marked as half of one.
2. **The whole run fails retryably**. The worker re-queues with exponential backoff (1s, 2s, 4s), up to the attempt's run limit of 3.
3. **Runs exhausted, or a non-retryable error**. The attempt lands in `failed` with the reason kept and the submission intact; the learner re-runs it from the UI with a fresh budget.

Evaluation runs off a `JobQueue<T>` interface with an in-process implementation. It is honest about being in-process: a restart loses queued jobs, leaving the attempt in `queued` for a re-run. That is an acceptable trade when the submission itself is already durable, and a durable broker is a dependency this MVP has not earned.

6. Extensibility: the four seams, and what each costs

Change	What you write	What you do not touch
A new submission format (image + OCR, repo link, JSON export)	One <code>SubmissionNormalizer</code> , registered in the container	Any evaluator, service, route, or the aggregator
A new evaluation approach (a static analyser over submitted code, a peer-review queue, a second model voting)	One <code>Evaluator</code> , added to the pipeline's optional list	The pipeline, the worker, the attempt lifecycle
A new problem	One entry in <code>seedProblems()</code> : signals, concepts, pitfalls, weights	No code at all
A real queue / database / model provider	One adapter behind <code>JobQueue</code> , <code>AttemptRepository</code> , <code>LlmClient</code>	One line each in <code>container.ts</code>

The composition root (`container.ts`) is the only file in the codebase that names a concrete adapter. Everything else takes its collaborators through its constructor.

7. Key trade-offs

Structured submission as the primary format. *Cost:* more friction than a text box, and it nudges toward a conventional shape. *Why anyway:* the friction is the teaching. A learner who has to write one line of responsibility per class discovers the overloaded class themselves, which is the lesson.

Signal matching rather than parsing meaning. Requirement coverage is keyword-set matching over the design text. *Cost:* it can be gamed by name-dropping, and it can miss a design that solves a requirement in unusual vocabulary. *Why anyway:* it is instant, free, reproducible, and it can show its evidence, which is what makes it arguable rather than authoritative. The model covers what it misses, and the UI shows exactly what matched so a false verdict is visible.

Regex-based code parsing rather than a real AST. *Cost:* it will misread exotic syntax. *Why anyway:* the evaluators need shape, not precision, and the model reads the raw source anyway. Three language parsers is a week of work for a sharper answer to a question already half-answered. `CodeNormalizer` is the single class to replace if code becomes the popular format.

In-process queue. *Cost:* jobs do not survive a restart. *Why anyway:* the assignment explicitly warns against turning this into a distributed-systems project, the submission is durable, and the interface means the upgrade is an adapter.

Attempts are immutable once submitted. *Cost:* a typo means starting a new attempt. *Why anyway:* the history is the product. An editable attempt makes "did I improve?" unanswerable.

No accounts. *Cost:* clearing browser storage loses your history. *Why anyway:* auth is orthogonal to everything being evaluated here, and the learner identity is isolated in one function on each side, so adding it is a small, contained change.

8. If it needed to handle more users or a slower model

Kept brief, per the brief's scope boundary. The two constrained resources are model latency/cost and evaluation throughput: nothing else in this system is heavy.

- **The queue interface is already the seam.** Swap `InProcessJobQueue` for BullMQ (Redis) or a Kafka topic and run the worker as a separate process. No domain code changes; the attempt lifecycle already assumes evaluation is asynchronous and can fail.
- **Cache the deterministic half.** It is a pure function of (submission, problem version), so identical resubmissions never re-run.
- **Polling** → **SSE** on the attempt endpoint when concurrent learners make the poll volume matter.
- **The deterministic floor is the load-shedding mechanism.** Under model pressure, stop scheduling the LLM evaluator and every learner still gets an immediate, honest, clearly-marked partial review instead of a queue that grows.

DesignDojo: an LLD practice platform

Practise Low-Level Design the way you practise algorithms: pick a problem, submit a design, get feedback that points at your actual classes, and try again with the last review still in view.

Built for the CipherSchools 2-day engineering assignment.

Research note	<code>docs/RESEARCH.md</code> : the learner problem, what exists, the gaps, and the product direction
Design note	<code>docs/DESIGN.md</code> : MVP, flow, classes, evaluation approach, trade-offs
AI usage	<code>AI_USAGE.md</code> : five decisions where AI changed the outcome, including the ones I rejected

Quick start

Requires **Node 20+**. No database and no API key needed.

```
npm install
npm run dev
```

- App → <http://localhost:5173>
- API → <http://localhost:4000/api/health>

That is the whole setup. With no `MONGODB_URI` the platform stores attempts in memory, and with no `ANTHROPIC_API_KEY` it uses an offline stand-in reviewer, labelled as such in the UI, so the full loop works on a fresh clone.

With the real reviewer model

```
cp .env.example .env
# add ANTHROPIC_API_KEY=sk-ant-...
npm run dev
```

The header badge switches from `offline reviewer` to the model id. To persist attempts, add `MONGODB_URI=mongodb://localhost:27017/lld-practice`; the badge then reads `mongodb`.

Tests

```
npm test          # 85 tests
npm run typecheck
```

The loop

DesignDojo

ProblemsProgress

offline reviewerin-memory

PROBLEMS / PARKING LOT / ATTEMPT #2

Parking Lot

evaluated

The brief

Design the object model for a multi-floor parking lot.

Vehicles arrive at an entrance, are assigned a suitable parking spot, and are issued a ticket. On the way out they pay based on how long they stayed and the kind of vehicle they brought, and the spot is released for the next vehicle.

The lot has several floors. Spots come in sizes (motorcycle, compact, large) and a

Requirements

1. A vehicle can be parked and receives a ticket that identifies its spot and entry time.

2. A vehicle is only assigned a spot that its size fits.

3. Spots are organised across multiple floors.

4. The exit fee depends on duration and vehicle type.

5. Leaving frees the spot for reuse.

6. Electric vehicles can be assigned a spot with a charger.

Timeline

Attempt started

just now

Submitted for review

88

/100

ATTEMPT #2 · ▲ 36 FROM LAST ATTEMPT

Your design puts 8 types on the problem, with PricingStrategy carrying the variation. The shape and the reasoning both hold up; push on the failure paths next.

deterministic · 21ms

llm-reviewer · 1ms

Rubric breakdown

Requirement Coverage100 · 18% weight

rules 100 · 6/6 stated and 3/3 implied requirements detected in the design.

Abstraction & Responsibility90 · 20% weight

rules 100 · reviewer 85: 5/5 reference concepts modelled as their own type, 8 types; 0 crowded; 8 with a stated responsibility. Based on the number of types and whether each has a single stated job.

Relationships & Cardinality90 · 15% weight

rules 100 · reviewer 80: 7 relationships, 6/6 with cardinality, 0 orphan types. Based on relationship count, direction and multiplicity.

Extensibility81 · 18% weight

rules 88 · reviewer 78: 3 interfaces, 0 polymorphic links, 3 policy-style collaborators. Based on whether varying behaviour sits behind an abstraction.

Edge Cases & Concurrency81 · 15% weight

rules 100 · reviewer 68: 3/3 unstated cases anticipated; 4 failure-path and 2 concurrency signals. Based on whether failure and contention paths appear in the design.

Trade-off Reasoning85 · 14% weight

rules 100 · reviewer 85: 133 words of notes with 6 comparison markers. Based on the depth of the written justification.

Feedback

Every stated requirement is accounted for

Strength

RULE

All 6 requirements map onto something concrete in your design.

Choose → design → submit → review → try again.

DesignDojo

Problems

Progress

offline feedback

in memory

PRACTICE

Pick a problem

Design it, submit it, and get feedback that points at your actual classes. Every attempt is kept, so you can see what changed when you try again.

Parking Lot

easy

40 min

best 88

2 attempts

Vending Machine

easy

30 min

best 37

1 attempt

Elevator System

hard

50 min

not attempted

Expense Splitter

medium

40 min

not attempted

Catalogue: four problems, your best score on each

DesignDojo

Problems

Progress

offline feedback

in memory

PROBLEMS

ELEVATOR SYSTEM

draft saved just now

The brief

Design the object model for the elevator system of an office building.

The building has several floors, and several elevator cars. A person on a floor presses up or down, a person inside a car presses a destination floor. The system decides which car serves which request and in what order, then the cars move, stop and open their doors.

Cars can be taken out of service for

Structured design

Code

Class diagram

Submit for review

Submit for review

Classes & interfaces

One line per type. The responsibility is what gets reviewed, if it needs an "and", you probably have two items.

Parkinglot

class

Aggregate root that owns the floors and coordinat

park(vehicle), unpark(ticket)

Parkingfloor

class

Holds the spots on one level and reports which an

findAvailable(size)

ParkingSpot

class

A single allocatable space with a size and current

size, currentTicket, hasCharger

Vehicle

class

The vehicle being parked, knowing its plate and si

licensePlate, size, isElectric

Relationships

Why does a reference to whom, and how many. Cardinality is graded.

Parkinglot

composition

Parkingfloor

1..*

Parkingfloor

composition

ParkingSpot

1..*

Requirements

1. A person on a floor can request a car by direction.

2. A person inside a car can request a destination floor.

3. The system chooses which car serves each request.

4. A car moves between floors, stops, and opens its doors.

5. A car serves its pending stops in a sensible order rather than strictly first-come.

6. A car can be out of service or restricted to certain floors.

Timeline

Attempt started

just now

Workspace: the brief beside the design, autosaved

DesignDojo

Problems

Progress

offline feedback

in memory

PROBLEMS

PARKING LOT

easy

40 min

best 88/100

The problem

Design the object model for a multi-floor parking lot.

Vehicles arrive at an entrance, are assigned a suitable parking spot, and are issued a ticket. On the way out they pay based on how long they stayed and the kind of vehicle they brought, and the spot is released for the next vehicle.

The lot has several floors. Spots come in sizes (motorcycle, compact, large) and a vehicle may only occupy a spot that fits it. Some spots are reserved for electric vehicles and have a charger attached.

Model the classes, their responsibilities, and the relationships between them. You are not being asked for a working implementation, a database schema, or an API, just the design.

How this is scored

Requirement Coverage

18%

Abstraction & Responsibility

20%

Relationships & Cardinality

15%

Extensibility

18%

Edge Cases & Concurrency

15%

Trade-off Reasoning

14%

Your attempts (2)

#2

88/100

just now

#1

52/100

just now

Brief: requirements, constraints, and how it is scored

DesignDojo

Problems

Progress

offline feedback

in memory

PROGRESS

Your attempts

3 attempts across 2 problems - average score 58

Vending Machine

1 attempt

#1

37/100

weakest: Trade-off Reasoning 15

just now - structured

Parking Lot

2 attempts - improved 36 points

#2

88/100

weakest: Extensibility 81

just now - structured

#1

52/100

weakest: Trade-off Reasoning 15

just now - structured

Progress: every attempt kept and charted

What makes the feedback useful

There is more than one correct design for a parking lot, so grading against a reference solution would be actively wrong. Three things fall out of taking that seriously.

1. Evaluation is split between rules and a model

Checkable things are decided by reproducible rules; judgement calls go to a language model. The rubric declares the blend **per dimension**:

Dimension	Rules	Model	Why
Requirement Coverage	70%	30%	A finite authored checklist: mostly a fact
Relationships & Cardinality	50%	50%	Presence is a fact; correctness is judgement
Edge Cases & Concurrency	40%	60%	Vocabulary is detectable; sufficiency is not
Abstraction & Responsibility	30%	70%	Crowded classes are detectable; good taste is not
Extensibility	30%	70%	Seams are detectable; the <i>right</i> seams are not
Trade-off Reasoning	0%	100%	Only a reader can judge an argument

The deterministic half is the platform's floor: dependency-free, single-digit milliseconds, identical every run. The model is given the rule findings and told not to repeat them, so the two compose into one review.

2. Every finding says where it came from

Each item is tagged `rule` or `reviewer` on screen. Requirement coverage shows the learner's own words that matched. A verdict you disagree with is something you can argue with, not something handed down, which matters a lot when the model is occasionally wrong.

3. Rules ask rather than assert

No check says a concept is missing. It says nothing was *detected* that owns it, and asks whether that was deliberate, with a score floor, so departing from the reference costs some marks and never all of them.

Concepts are graded three ways rather than two: **dedicated** (a type is named for it), **folded** (it exists only as a method on another class, half credit and a question), or **absent**. Full credit for folding it in would let name-dropping score as well as modelling; no credit would assert that one decomposition is the only right one.

When evaluation is slow or fails

Evaluation is asynchronous, so submitting returns immediately with the attempt `queued` and the client polls. Three levels of failure handling, and at none of them is the learner's work lost:

1. **The model fails or returns unparseable JSON** → the review completes from the deterministic checks alone, flagged `degraded` with the reason, and the UI shows a partial-review banner with a re-run button.
 2. **The whole run fails retryably** → re-queued with exponential backoff (1s, 2s, 4s), up to 3 runs.
 3. **Runs exhausted** → the attempt lands in `failed` with the reason kept and the submission intact; the learner re-runs it with a fresh budget.
-

Architecture

web/	React + Vite, no UI framework
server/src/	
domain/	← no imports from anything below this line
problem/	Problem, Rubric (dimensions + deterministic share)
attempt/	Attempt aggregate, state machine, Submission union, DesignModel
evaluation/	Evaluation value objects
evaluation/	
normalizers/	Structured Code Diagram → DesignModel (Strategy)
deterministic/checks/	8 independent DesignChecks
llm/	LlmClient (Anthropic offline), prompt, schema-validated parse
EvaluationPipeline.ts	composes evaluators, degrades gracefully
ScoreAggregator.ts	blends by the rubric's deterministic share
application/	AttemptService, EvaluationWorker, JobQueue port
infrastructure/	repositories (in-memory Mongo), HTTP, seeded problems
container.ts	the only file that names a concrete adapter

Dependencies point inward. The domain imports nothing from `infrastructure` , which is what lets the whole suite run against in-memory adapters in about a second.

Extending it

To add...	You write	You do not touch
A submission format	one <code>SubmissionNormalizer</code>	any evaluator, service or route
An evaluation approach	one <code>Evaluator</code>	the pipeline, worker or lifecycle
A problem	one entry in <code>seedProblems()</code>	no code at all
A real queue / DB / model provider	one adapter	one line in <code>container.ts</code>

API

All routes are under `/api` . The learner is identified by an `x-learner-id` header (see *Limitations*).

Method	Route	Purpose
GET	<code>/health</code>	Storage and reviewer actually in use, queue depth
GET	<code>/problems</code>	Catalogue
GET	<code>/problems/:idOrSlug</code>	Brief, requirements, constraints, rubric weights
POST	<code>/attempts</code>	Start an attempt ({ problemId })
PUT	<code>/attempts/:id/draft</code>	Autosave
POST	<code>/attempts/:id/submit</code>	Submit; returns immediately as <code>queued</code>
GET	<code>/attempts/:id</code>	Poll for status and the evaluation
POST	<code>/attempts/:id/reevaluate</code>	Re-run a failed evaluation
GET	<code>/attempts?problemId=</code>	History

The grading key (requirement signals, reference concepts, pitfalls) never leaves the server; there is a test asserting it.

Tests

85 tests, ~1.5s, no external services.

File	Covers
<code>attempt-lifecycle.test.ts</code>	State machine: happy path, illegal transitions, retry budget, and that a rejected submission leaves the attempt editable
<code>evaluation-pipeline.test.ts</code>	Strong vs weak scoring separation, that the weak design is blamed for the <i>right</i> dimensions, reproducibility, the rule/model blend, and four degradation paths (model down, prose instead of JSON, wrong schema, fenced JSON)
<code>evaluation-worker.test.ts</code>	Retry with backoff, exhausting the budget into <code>failed</code> , learner-triggered re-run, redelivered jobs, previous-score context
<code>normalizers.test.ts</code>	TypeScript / Java / Python parsing, Mermaid arrow direction and cardinality, unparseable input, unregistered format
<code>scoring.test.ts</code>	Rubric normalisation and validation, blending, renormalising over assessed dimensions, clamping, JSON extraction edge cases
<code>api.test.ts</code>	End-to-end loop, repeat attempts showing improvement, cross-learner isolation, 422/409/404 paths, running with no reviewer at all
<code>problem-catalogue.test.ts</code>	Every seeded problem is completely authored, its rubric covers all six dimensions, its signals are actually matchable, and it separates a strong design from a generic one: a guard that covers each problem added later with no new fixture
<code>signal-matching.test.ts</code>	Word-boundary and inflection rules, camelCase/snake_case identifier splitting, and that a dedicated type scores above the same concept folded into a god class

Limitations

Honest about what this MVP is not:

- **No accounts.** A per-browser id in `localStorage`; clearing it loses your history. Isolated to one function on each side.
- **In-process queue.** A restart loses queued jobs, leaving the attempt in `queued` for a re-run. The `JobQueue` interface is the upgrade path.
- **Requirement coverage is signal matching, not comprehension.** It can be gamed by name-dropping and can miss unusual vocabulary. It shows its evidence precisely so you can see when it is wrong.
- **Code parsing is regex-based**, not an AST. It reads shape, not semantics; the model reads the raw source anyway.
- **The offline reviewer is not a model.** It is a heuristic stand-in so the loop works without a key, and it says so in the UI and in the evaluation's provenance.

- **Four problems**, seeded in code. Adding more is a content task, not an engineering one.

AI usage

I used Claude (via Claude Code) as a pair programmer on this build. The working loop was the same every time: I decided what the thing needed to do, the AI argued a position and drafted faster than I type, and nothing stayed in the repo until I had read it and a test had run against it.

That loop is why this document is mostly about disagreements. The AI is a useful second opinion precisely because it will confidently propose something reasonable-but-wrong, and the value is in catching which is which. Below are the five exchanges that actually changed the outcome, including the three where I went the other way.

1. Rejected: "let the LLM do the whole evaluation"

Suggested. The first architecture I was offered was the obvious one: normalise the submission, send everything to the model with a rubric in the prompt, parse a score back. Simple, and it covers every dimension in one call.

I rejected it, and this became the central design decision instead.

Two reasons, both from actually testing the behaviour. First, **it is unreliable at the checkable things**. Asked "does this design address the six stated requirements", the model missed a genuinely-absent requirement on some runs and hallucinated coverage on others, while a keyword-set match over the design text answers that exactly, instantly, and for free. Second, **it is not reproducible**. Resubmitting the same design produced scores several points apart. A platform whose entire premise is "attempt 2 scored higher than attempt 1" cannot have that much noise in its instrument.

What I built instead. A two-part pipeline where the rubric declares a per-dimension `deterministicShare`: requirement coverage is 70% rules, trade-off reasoning is 100% model. The deterministic half is the floor: dependency-free, milliseconds, reproducible, and able to show its evidence. The model handles what only a reader can judge.

The knock-on benefit is the one I did not anticipate: because the deterministic half cannot really fail, **a model outage degrades the review instead of breaking it**. That fell out of the split rather than being designed for.

2. Accepted, then had to correct: the "more than one valid design" problem

Suggested. When I asked how to give feedback when several designs are valid, the answer was to prompt the model with "there is no single correct solution; judge on internal consistency". Correct as far as it goes, and it is in the system prompt.

Where it was incomplete. It only addressed the model. My *deterministic* checks had exactly the same bias baked in, and worse, because rules are absolute: `ConceptCoverageCheck` compares against a reference concept list, so a learner who deliberately folded pricing into `Ticket` would be marked down by a rule with no room to disagree. The AI did not flag this, presumably because I had asked about the prompt.

What I changed. Two things, both in the deterministic half:

- **Wording.** No check says a concept is missing. It says nothing was *detected* that owns it, and asks whether that was deliberate: *"A common approach is a dedicated abstraction, but folding it into an existing class can be the right call, if you did that deliberately, say which class owns it and why."*
- **Score floors.** `ConceptCoverageCheck` floors at 40, so departing from the reference costs some marks and never all of them. `RequirementCoverageCheck`, where there genuinely is a right answer, has no floor.

And the reason I trust it now: **every finding is labelled with its source in the UI**, and requirement coverage shows the learner's own matched words. A verdict the learner disagrees with is arguable rather than authoritative.

3. A test written by AI found a real bug in my domain logic

I asked for edge-case tests around the attempt lifecycle rather than writing only the happy paths myself. One asserted something I had not thought about: after a submission is rejected for being too thin, *what state is the attempt in?*

It failed. My `submit()` was:

```
this.transitionTo('queued');    // ← state already changed
assertSubmittable(submission);  // ← then throws
```

The transition happened before validation, so a learner whose submission was rejected would have found their attempt stuck in `queued`: uneditable, with nothing queued, and their work effectively frozen. Reversed to validate first, so a rejected submission leaves the attempt exactly as it was. The test is `attempt-lifecycle.test.ts` > `rejects a submission too thin to be worth evaluating`, and it asserts the attempt is still `draft` afterwards.

This is the clearest value I got from AI on this project: not the code, the *questions*. I would have tested that submitting invalid input throws. I would not have tested what it leaves behind.

4. Prompt design: three rules that exist because of observed misbehaviour

The reviewer system prompt was written iteratively against real outputs, not in one pass. Three of its five rules were added to fix specific failure modes:

Behaviour without the rule	Rule added
Graded by resemblance to the reference concepts, marking down unconventional but sound designs	"Judge the design on its own internal logic... never on how closely it resembles a reference solution."
Produced fluent, generic advice that could have been written without reading the submission	"Every point must quote or name something specific from their submission. If you cannot point at their words, do not make the point."
Re-checked requirement coverage, duplicating the rules and occasionally contradicting them	"Do not comment on whether the stated requirements are covered... its findings are given to you below. Do not repeat or contradict them."

Two more settings came from the same testing: **temperature 0.1**, because two identical designs getting different verdicts destroys the progress history the product is built on; and passing the **deterministic findings into the prompt**, which is what turns two evaluators into one coherent review.

I also rejected the suggestion to describe scoring as "against a perfect design". It produced uniformly harsh scores, which is demotivating and inaccurate for a 45-minute exercise. The prompt now says *"against what a strong candidate would produce in a 45-minute interview... 50 is an average attempt."*

5. Accepted with a change: the intermediate representation

Suggested. For multi-format support, one evaluator per format: `StructuredEvaluator`, `CodeEvaluator`, `DiagramEvaluator`.

Why I changed it. That multiplies: n formats $\times$ m evaluators, and every new rule has to be written three times and kept consistent. I inverted it: normalise all three formats into one `DesignModel`, and let every evaluator read only that. Formats and evaluators become independent axes: a new format is one normaliser and zero evaluator changes; a new evaluator works across all formats immediately.

The suggestion was reasonable and would have worked for three formats. It just would not have survived the fourth.

How I kept myself honest

Nothing went in on the strength of looking right. Four things in this repo exist because the verification step caught something:

- **The `submit()` ordering bug.** A lifecycle test asserted what state an attempt is left in after a submission is *rejected*. It failed: the transition to `queued` ran before validation, so a learner whose submission was too thin would have found their work frozen in a state they could not edit. Fixed by validating first.
- **A regex that silently shifted every capture group.** A cleanup pass collapsed `[, , generic, arrayOf, plain]` into `[, generic, arrayOf, plain]` in `CodeNormalizer`. Nothing looked wrong; two tests went red immediately and named it.
- **Three scoring flaws, found by using my own product.** I wrote realistic designs for all four problems by hand and ran them through the platform. The verdicts were not convincing: concept coverage counted a mention as ownership, so name-dropping `calculateFee()` scored as well as modelling a `PricingStrategy`; the pitfall check returned 100 when it found nothing, quietly lifting every abstraction score; and edge-case detection used a global word list that missed designs handling the cases in their own vocabulary. All three are now fixed, and §2 below is the philosophy that came out of it.
- **Test thresholds that were calibration, not invariants.** When retuning broke three tests, two turned out to assert arbitrary numbers. I rewrote them to assert the actual invariant (requirement coverage must equal the deterministic score when the model owns none of it) and added one asserting the weak design is blamed for the *right* dimensions, because a score that is correct by accident is not feedback.

The division of labour

I pointed, reviewed and corrected; the AI drafted. It was quickest on work where the shape was already settled: the Express/Vite/Vitest scaffolding, the repetitive DTO and Zod schema mapping, the CSS design system, and the Mermaid arrow-parsing table, which is fiddly and exactly the kind of thing a model gets right faster than a person does.

None of it landed unedited. The generated requirement signal lists were the clearest case: the synonym sets were too narrow and missed naming a real learner would reach for, like `RateCard` for pricing, so I rewrote them and then added the `problem-catalogue` test suite to stop the same narrowness creeping into any problem added later.

Where the judgement had to be mine

Three calls carry this project, and no amount of drafting speed would have produced them:

1. **Splitting evaluation by dimension rather than by evaluator** (§1). The rubric's `deterministicShare` is the whole design in one number: requirement coverage is 70% rules because it is a checklist, trade-off reasoning is 100% model because only a reader can judge an argument.
2. **Treating "more than one valid design" as a scoring problem, not a prompting one** (§2). The prompt fix was the easy half; the rules had the same bias and rules are absolute, which is why concepts are graded dedicated / folded / absent with a floor.
3. **Deciding what a learner must hand in**. The structured format exists because free prose lets someone write three fluent paragraphs while never naming a responsibility or a cardinality, which are the two things being taught. The friction is the teaching.

I used the AI to attack all three after the fact, which is where §2 and §3 came from. It is a good adversary and a poor author: it will tell you a design is fine, and it will also find the hole in it if you ask it to try.